

Lecture	Content	Sipser Book	Quiz
Midterm Exam Syllabus			
Lecture 1	Logistics, Introduction to automata	Sipser 0, 1.1	
Lecture 2	Formal definition of Deterministic Finite Automata (DFA)	Sipser 1.1	
Lecture 3	Problem Solving on Designing DFA.		
Lecture 4	Regular languages, closure under: complementation, union		
Lecture 5	Formal definition of Nondeterministic Finite Automata (NFA)	Sipser 1.2	
Lecture 6	Equivalence of DFAs and NFAs		Quiz 1
Lecture 7	Introduction to regular expression	Sipser 1.3	
Lecture 8	More examples on Regular expression		
Lecture 9	Equivalence of DFAs and regular expressions		Quiz 2
Final Exam Syllabus			
Lecture 10	Limits of regular languages: the pumping lemma	Sipser 1.4	
Lecture 11	More examples of pumping lemma	Sipser 1.4	
Lecture 12	Introduction to Context Free Grammar (CFG)	Sipser 2.1	
Lecture 13	Context Free Grammar (CFG) examples		
Lecture 14	Parse Trees, Derivations, and Ambiguities.		
Lecture 15	Push Down Automata (PDA)	Sipser 2.2	Quiz 3
Lecture 16	More on PDA		
Lecture 17	Introduction to Turing machines (TM)	Sipser 3.1	
Lecture 19	More on models, encodings of inputs and proving decidability	Sipser 4.1	Quiz 4
Lecture 20	Proving undecidability by diagonalization	Sipser 4.2	